Scalable fit of the output-error sensitivity of a Clements unitary mesh vs. splitting-ratio error

FIT on the small range $[0, 0.1]$

Model (overlap-compatible, scalable in N/N_0): $E_S(\sigma_S, N) = \sqrt{2}\,\sqrt{1 - e^{-u_S^2}}$, $u_S = \dfrac{s_S(N)\,\sigma_S}{\sqrt{2}}$, $s_S(N) = s_{S,0}\,(N/N_0)^{p_S}$, $N_0 = 100$

$s_{S,0}$ = initial slope $dE/d\sigma_S$ at the reference size $N_0 = 100$ (dimensionless)

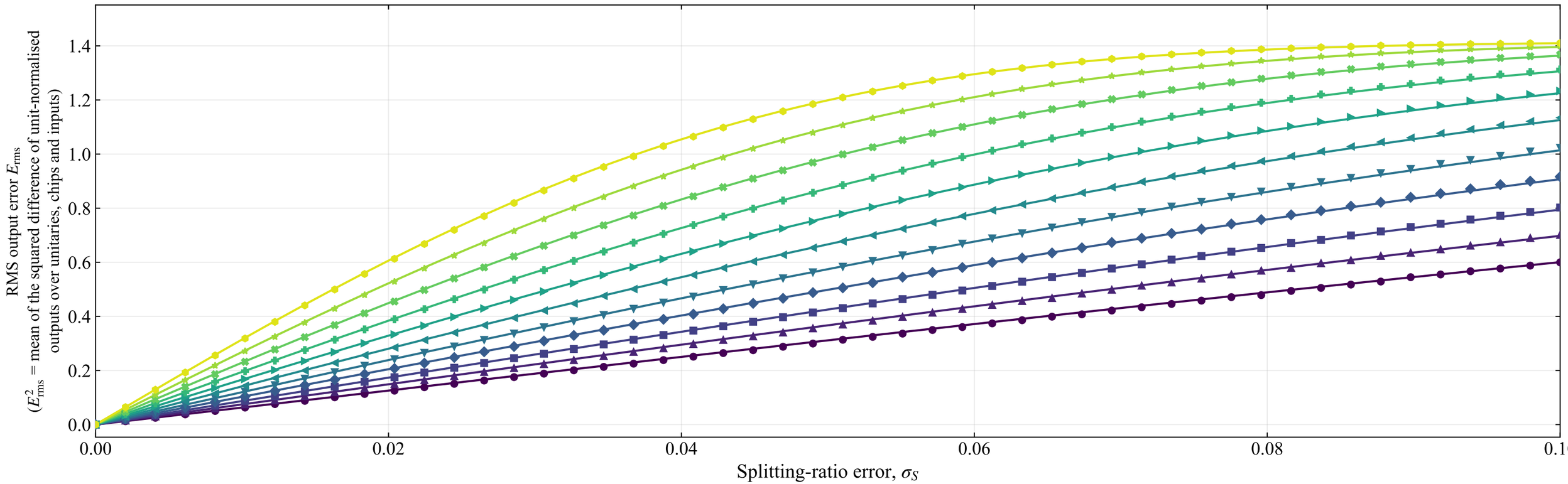
p_S = network-size scaling exponent of the sensitivity ($p_S \approx 0.5$: $\sqrt{N}$ growth)

Best-fit values (small range, all N, equal weight per point; ± = bootstrap s.d. over unitaries): $s_{S,0} = 14.17384 \pm 0.00091$, $p_S = 0.50325 \pm 0.00006$

Fit quality (fit range): all 550 points: RMSE = 0.00317, MAE = 0.00253, $R^2 = 0.99994$, max|ΔE| = 0.0099 | off-plateau ($E \le 1.2$, 475 points): RMSE = 0.00317, $R^2 = 0.99991$, mean rel. err. = 0.67%

Hardware model: coupler splitting ratio $\eta \sim \mathrm{TruncNorm}(1/2, \sigma_S^2; 0, 1)$ for both couplers of every MZI; all other channels ideal | markers = simulated E_{rms} (over unitaries, chips and input types), solid lines = scalable fit (fit in E space)

$N = \{20, 28, 38, 53, 72, 100, 138, 190, 263, 362, 500\}$ | Haar-random unitary U (single Clements mesh) | 7 input types × 10 draws, pooled (equal weight) | one forward pass $y = Ux$ | $E_{\mathrm{rms}} = \sqrt{\langle \|\hat{y}^{\mathrm{hw}} - \hat{y}^{\mathrm{ideal}}\|_2^2 \rangle}$ | 25 unitaries × 30 chips | 50 points per curve



Network size N | scaled sensitivity from the fit (markers: data, lines: fit)

$N = 20$, $s_S = 6.306$	$N = 53$, $s_S = 10.297$	$N = 138$, $s_S = 16.668$	$N = 362$, $s_S = 27.081$
$N = 28$, $s_S = 7.469$	$N = 72$, $s_S = 12.014$	$N = 190$, $s_S = 19.578$	$N = 500$, $s_S = 31.860$
$N = 38$, $s_S = 8.710$	$N = 100$, $s_S = 14.174$	$N = 263$, $s_S = 23.058$	